

Problem Set 5 – Calibration Log Synthesizer

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Source Code

```
1  # For a batch of length batch_length (ID at index 0), i counts up from
2  # 1 to batch_length - 1, and target_index = batch_length - i maps that
3  # to a right-to-left position. At i=1, target_index is the last index;
4  # as i grows it shrinks down to 1, the first reading after the ID.
5  #
6  # The while loop stops on its own because slicing past the end of a
7  # list just returns [] instead of erroring. Once feed_cursor passes the
8  # last index, the one-item slice is empty (falsy), and the loop ends -
9  # no separate length check needed.
10 #
11 # Author: Kshitiz
12
13 calibration_feed = [
14     [201, 6.0, 9.5, "IGNORE", 4.0],
15     [],
16     [202, 11.2, "FAULT", 7.8, 5.5],
17     [203, 14.0, 3.5, 8.25],
18     [204, 2.75, "HALT", 6.0],
19 ]
20
21 feed_cursor = 0
22 total_valid_readings = 0
23 global_max = None
24 global_min = None
25 checksum = 0.0
26 emergency_stop = False
27
28 while (current_slice := calibration_feed[feed_cursor:feed_cursor + 1]):
29     batch = current_slice[0]
30
31     if not batch:
32         print(f"Batch {feed_cursor} is EMPTY. Proceeding.")
33         feed_cursor += 1
34         continue
35
36     calibration_id = batch[0]
37     print(f"Evaluating Batch {feed_cursor} (ID: {calibration_id})...")
38
39     # Manual length count (no len()).
40     batch_length = 0
41     for _ in batch:
42         batch_length += 1
43
44     batch_sum = 0.0
45
46     for i in range(1, batch_length):
47         target_index = batch_length - i
48         reading = batch[target_index]
49
50         if reading == "IGNORE":
51             print(f"Signal IGNORE encountered at Batch {calibration_id}.")
```

```

52         continue
53
54     if reading == "FAULT":
55         print(f"Signal FAULT detected. Suppressing batch {calibration_id}.")
56         break
57
58     if reading == "HALT":
59         print("Signal HALT detected. Executing emergency protocol.")
60         emergency_stop = True
61         break
62
63     batch_sum += reading
64     total_valid_readings += 1
65
66     if global_max is None or reading > global_max:
67         global_max = reading
68     if global_min is None or reading < global_min:
69         global_min = reading
70
71     else:
72         if calibration_id % 2 == 0:
73             batch_sum *= 1.5
74         else:
75             batch_sum *= 0.8
76         checksum += batch_sum
77
78     if emergency_stop:
79         break
80
81     feed_cursor += 1
82
83 print()
84 print("=" * 40)
85 if emergency_stop:
86     print("CALIBRATION COMPLETE: EMERGENCY TERMINATION")
87 else:
88     print("CALIBRATION COMPLETE: FULL FEED PROCESSED")
89 print("=" * 40)
90 print(f"Total Valid Readings Processed: {total_valid_readings}")
91 print(f"Global Calibration Checksum: {round(checksum, 2)}")
92 print(f"Maximum Reading Encountered: {global_max}")
93 print(f"Minimum Reading Encountered: {global_min}")
94
95 # Submission Note: Batch 2 (FAULT) and Batch 4 (HALT) both had their
96 # else block skipped since break was hit in the inner loop - their
97 # valid readings still count toward the total, but their sums never
98 # reach the checksum. Batches 0 and 3 finished without breaking, so
99 # their else blocks ran and got added to the checksum.

```

Trace Log

```

1 Evaluating Batch 0 (ID: 201)...
2 Signal IGNORE encountered at Batch 201.
3 Batch 1 is EMPTY. Proceeding.
4 Evaluating Batch 2 (ID: 202)...
5 Signal FAULT detected. Suppressing batch 202.
6 Evaluating Batch 3 (ID: 203)...
7 Evaluating Batch 4 (ID: 204)...
8 Signal HALT detected. Executing emergency protocol.
9
10 =====
11 CALIBRATION COMPLETE: EMERGENCY TERMINATION

```

```
12 =====
13 Total Valid Readings Processed: 9
14 Global Calibration Checksum: 36.2
15 Maximum Reading Encountered: 14.0
16 Minimum Reading Encountered: 3.5
```

Output matches the worked trace in the assignment exactly, including the checksum of 36.2, max of 14.0, and min of 3.5.

Submission Note

Batch 2 (ID 202) and Batch 4 (ID 204) both had their inner loop's `else` block skipped. Batch 2 hit `FAULT` partway through, which breaks the inner loop, so its `else` never runs even though its two valid readings (5.5 and 7.8) already counted toward Total Valid Readings. Batch 4 hit `HALT` after one valid reading (6.0), which also breaks the inner loop, on top of setting the emergency flag that stops the whole feed. Batches 0 and 3 completed their inner loop without hitting `break`, so their `else` blocks ran normally and their weighted sums were added to the checksum.